

## HR Analytics Project – EDA Summary

**Project Objective** - Analyse workforce attrition drivers using a synthetic enterprise HR dataset (~15,000 records) built with business-driven relationships.

### Methodology

1. Synthetic dataset generation with business logic
2. Data quality issue injection (duplicates, nulls, label inconsistencies)
3. Cleaning pipeline and root cause remediation
4. Data dictionary creation
5. Hypothesis-driven EDA

### Key EDA Findings

- *Overall Attrition*- Attrition Rate: 13.57%- Workforce generally stable but retention challenges exist.
- *Compensation Analysis*-
  - Monthly Salary: Lower salary linked with higher attrition.
  - Salary Hike %: No meaningful impact.
  - Bonus: Lower bonus associated with higher attrition.
  - Stock Options: No significant relationship.
- *Career Growth Analysis*-
  - Promotion: Employees without promotion showed higher attrition.
  - Years in Current Role: Minimal impact.
  - Training Hours: No meaningful relationship.
  - Employment Type: No meaningful relationship.
- *Work Environment Analysis*-
  - Overtime Hours: Strongest attrition driver.
  - Work Life Balance: Strong inverse relationship with attrition.
  - Remote Work Days: Lower flexibility linked with attrition.
- *Employee Sentiment Analysis*-
  - Job Satisfaction: Strong predictor of attrition.
  - Engagement Score: Very strong predictor.
  - Absenteeism: Moderate early warning indicator.
- *Performance Analysis*-
  - Manager Rating: Strong relationship with attrition.
  - Performance Rating: No meaningful relationship.
  - Projects Assigned: Minimal impact.
- *Risk Validation*-
  - High Risk Employees: 37.1% attrition
  - Low Risk Employees: 0% attrition

### **Final Business Insights**

1. Burnout and excessive overtime are primary attrition drivers.
2. Employee wellbeing strongly influences retention.
3. Compensation competitiveness matters more than annual salary hikes.
4. Manager quality significantly impacts employee retention.
5. Employee satisfaction and engagement are strongest behavioural indicators.
6. Rule-based risk scoring can support proactive retention interventions

### **Business Suggestions**

Workforce attrition within the organisation appears to be driven primarily by workplace environment and employee experience factors rather than employee productivity metrics. Excessive overtime burden, poor work-life balance, limited workplace flexibility, weaker managerial effectiveness, low employee satisfaction and declining organisational engagement emerge as the strongest workforce retention challenges.

The organisation should prioritize employee wellbeing initiatives, strengthen managerial effectiveness programs and implement proactive employee risk monitoring systems to reduce workforce attrition and improve long-term employee retention.